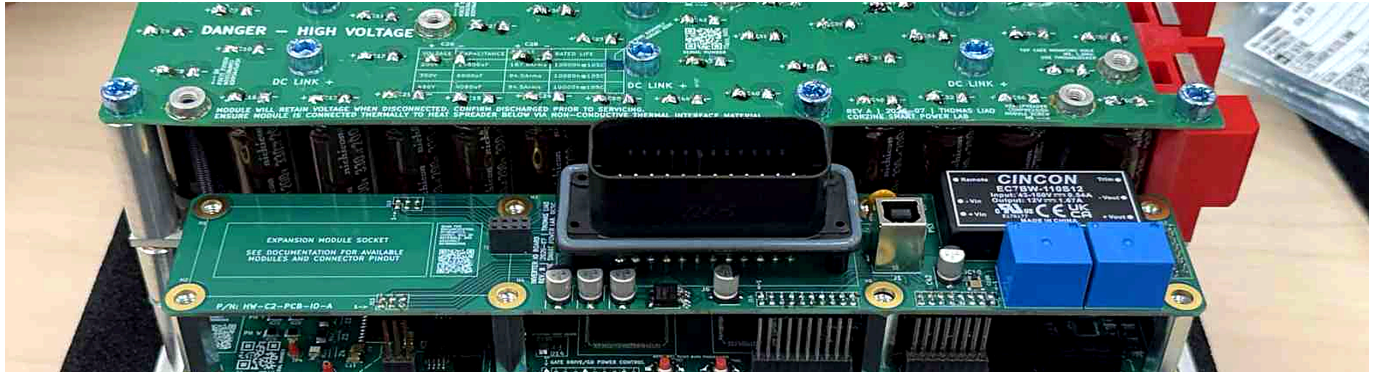


One control core. Power conversion at every scale.

OpenVVF separates a reusable control and safety core from the power hardware. The same interfaces support smaller user-supplied stages, the implemented Size 2 inverter, the released Size 2 DC/DC module, and future higher-power chassis.



Size 2 is a 200–450 V-class design rated 465 A continuous and 600 A peak for 60 seconds, with self-commissioning for PMSM and induction motors.

■ Modular by design

The controller is largely **power-stage agnostic**. PWM and protection interfaces can connect to lower-power or user-supplied gate-drive stages with appropriate sensing and gate-drive scaling, while the same core extends to higher-voltage and higher-current hardware.

■ Built and bench-tested

Size 2 has completed a **180 V bring-up**, a **one-hour endurance run**, and a **20-minute reversal stress test**. The safety architecture provides **six independent paths to the safe state**.

■ Open research at UCSC

OpenVVF is developed in the **Smart Power Lab** at UC Santa Cruz, with Professors **Keith Corzine** and **Leila Parsa** as faculty contacts. Hardware, firmware, safety documentation, and assembly data are public at docs.openvvf.org.

MODULAR POWER BLOCKS

Lower-power custom stages through Size 3

CONTROL MODULE Dual-MCU · FOC motor control + independent safety coprocessor · USB-B, dual CAN FD, and expansion I/O · RTE Studio + Telemetry Viewer



Lower-power stages

Interface-ready for smaller or user-supplied gate-drive hardware

■ SCALABLE INTERFACE



Size 2 inverter

Implemented reference power stage

■ IMPLEMENTED, UNDER TEST



Size 2 DC/DC

15 kW · 450 V class · reuses Size 2 boards and power stage

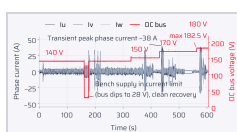
■ RELEASED DESIGN



Size 3

Up to 1,200 V · 1,400 A · modular liquid-cooled enclosure

■ IN DEVELOPMENT



Live telemetry, not promises

Ten published reports include raw, interactive logs for the browser-based Telemetry Viewer. Scan to open the 180 V bring-up: approximately 38 A peak current and clean recovery from a supply current-limit event.



SUPPORT THE SMART POWER LAB

Faculty and university contacts

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CURRENT SUPPORTERS

